Imputation of Missing Chemical Concentration and CAS Registry Numbers Using MICE

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AAI 510 – Machine Learning Fundamentals

Business Problem

- Incomplete CAS Numbers and concentration values reduce regulatory compliance, hazard assessment quality, and chemical inventory accuracy.

Research Objectives

- Evaluate MICE for missing value recovery,
- Compare joint vs concentration-only imputation,
- Identify limitations of high-cardinality categorical targets.

Dataset Overview

- 16,703 chemical product records
- Features: Trade Name, Chemical Name, CAS Number, Concentration
- High-cardinality dataset with 1,800+ CAS values.

Methodology

- Train/Test split (80/20)
- 20% masking in test data
- Ordinal encoding
- MICE (Iterative Imputer)
- 50 iterations
- Controlled evaluation protocol.

EDA & Data Preparation

Schema validation,

Completeness checks,

Categorical encoding,

Concentration distribution analysis,

Train-test segregation.

Model & Evaluation

- IterativeImputer (MICE)
- Metrics: Accuracy and Macro F1
- Evaluation only on masked values to ensure unbiased assessment.

Results

- CAS Number Accuracy: 0%
- Concentration Accuracy: 0.75%
- Macro F1: 0.20–0.24%
- Concentration prediction outperformed CAS prediction.

Business Impact & Deployment

Batch processing workflow for chemical registries.

Can improve data quality audits, compliance reviews, and downstream analytics.

Deploy via API or scheduled batch pipeline.

Conclusions & Future Work

- MICE provides a reproducible baseline.
- High-cardinality identifiers remain challenging.
- Future work: Random Forest, CatBoost, MissForest, Deep Learning approaches.